

ULTRASONIC WELDING FOR PLASTIC PARTS

A Comprehensive White Paper on Design, Manufacturing, Quality and Equipment

Scope:

Working principle • Machine architecture • Joint design • DFM rules • Process tuning • Materials • Quality control • Standards • Failure modes • OEM landscape • Comparison vs. competing methods

Audience:

Mechanical / electromechanical / product / manufacturing engineers and process specialists

Document length is intentional — it is meant to be a single, self-contained reference.

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1. Executive Summary

Ultrasonic welding is a solid-state plastic-joining process that converts high-frequency electrical energy into mechanical vibrations and applies them — under controlled compressive force — to the interface between two thermoplastic parts. Surface and intermolecular friction at the interface raises temperature locally, the polymer softens or melts, the parts collapse together over a controlled distance, and after a brief hold under pressure the joint solidifies. A typical weld cycle takes between 0.1 and 1.5 seconds, requires no consumables, no heat-up time, no adhesives and no fasteners, and leaves a hermetically sealed, high-strength bond. These attributes make it the dominant assembly process for high-volume thermoplastic parts in automotive interiors, medical disposables, consumer electronics housings, packaging, white goods and toys.

Despite this maturity, ultrasonic welding is unforgiving of poor part design and poor process control. Weld quality is determined far more by the geometry molded into the parts (the energy director, joint type, wall thickness and gating) and by the polymer chosen, than by the welder itself. A correctly designed butt joint with an energy director welded on a 20 kHz machine will outperform an ill-conceived joint on the most expensive servo-driven generator on the market. This paper is structured to give the reader exactly that emphasis.

The document is organized in seventeen sections. Sections 1 to 4 build foundations: what ultrasonic welding is, the physics behind it, the architecture of the machine and tooling, and the materials it can and cannot process. Sections 5 to 8 are the design core: joint geometries, design-for-manufacturing rules, dimensional tolerances and process parameters with their tuning logic. Sections 9 to 11 cover the production environment: setup, fixtures, horns and quality inspection. Sections 12 to 15 cover what governs and what threatens the process: standards, failure modes, common mistakes and the equipment landscape. Section 16 places ultrasonic welding against its competitors — vibration welding, laser welding, hot-plate welding, infrared welding, heat staking, solvent bonding, adhesives and mechanical fasteners — with a tabular guide to which method is best in which situation. Section 17 lists primary references for further study.

Bottom line up front

Ultrasonic welding is fast, clean, repeatable and consumable-free, but its results are dictated by the joint that has been molded into the part long before the part reaches the welder. Get the energy director, the wall thickness and the polymer right, and welding becomes a matter of selecting the right control mode (energy or distance) and tuning three numbers — amplitude, force and weld limit. Get them wrong, and no amount of process tuning will rescue the joint.

2. Introduction and Overview

2.1 What is ultrasonic welding?

Ultrasonic welding is a class of vibration welding in which mechanical oscillations above the audible range — typically 15 kHz to 70 kHz — are applied perpendicular to the joint plane between two thermoplastic parts held under compressive force. The oscillations cause cyclic compression and shear deformation across the interface; the work done against internal damping and surface friction generates heat at the joint, polymer chains become mobile, the two surfaces fuse, and the assembly solidifies as the vibrations stop. The horn (the tool that contacts the upper part) does not heat up itself, and the bulk of each part remains close to ambient temperature, which is why the technique is sometimes loosely described as a "cold weld".

The process is a sub-family of friction welding. Spin welding, linear vibration welding, orbital vibration welding and ultrasonic welding all share the same underlying physics — frictional heat generation at a thermoplastic interface — but differ in how the relative motion is produced. Ultrasonic welding is unique in that the relative motion is produced by very small (10–125 μm) but very high-frequency oscillations, so the joint can be small and intricate, the cycle time can be very short, and the energy delivery is highly localized.

2.2 Brief history

The piezoelectric effect was discovered by the Curies in 1880, and the use of high-frequency mechanical waves for material processing emerged in the 1940s and 1950s, initially for metal welding and machining. The breakthrough for plastics came in the 1960s, when Robert Soloff at Branson Instruments observed that the woven plastic ribbons of a doll's bow could be bonded with the company's ultrasonic probe, and showed that rigid thermoplastic parts could be joined the same way if their geometry directed the vibration energy to a small triangular ridge at the interface. The triangular ridge — the energy director — and the basic 20 kHz pneumatic press architecture date from this period and remain the dominant joint feature and machine architecture today.

2.3 Why it is used

Ultrasonic welding is chosen when the priorities are some combination of: very short cycle time (often <1 second), low per-piece cost (no consumables), good cosmetic appearance, hermetic sealing, repeatable joint strength and freedom from fasteners or adhesives that would otherwise have to be inventoried, dispensed and cured. Typical applications include:

- Automotive: instrument cluster lenses, head-light housings, interior trim, air-bag assemblies, fluid reservoirs, filters.

- Medical and life sciences: blood-filter housings, IV components, diagnostic cartridges, dialyzers, catheter hubs, blister-pack tear-seals.
- Consumer electronics and white goods: switches, sensor housings, remote controls, kitchen-appliance bodies, vacuum-cleaner parts.
- Packaging: blister packs, tube-end seals, dairy and beverage closures, child-resistant lids.
- Textiles and nonwovens: continuous bonding of disposables (masks, gowns, hygiene products) at frequencies of 20–35 kHz.
- Toys, stationery and small consumer goods: housings, battery-compartment covers, wheels, structural fasteners.

2.4 Where it is not used

Ultrasonic welding cannot weld thermosets (epoxy, phenolic, melamine), cured rubber and most elastomers — these materials degrade rather than soften when heated. It also struggles with very large planar joints (above roughly 250 × 250 mm at 20 kHz) because a single horn cannot deliver uniform amplitude across that area; for these, vibration or hot-plate welding are usually preferred. It is generally unsuitable for components with sensitive internal electronics directly beneath the horn, with heavy mineral or glass-fibre filling above 30 % by weight, and for materials with very different melt temperatures (>20 °C apart for amorphous-to-amorphous, more for semi-crystalline pairs).

3. Working Principle and Physics

3.1 Energy conversion chain

Every ultrasonic weld is the product of four sequential energy conversions:

1. Mains AC (50 / 60 Hz, 110–480 V) is rectified and inverted by the generator to high-frequency AC at the operating frequency of the stack — most commonly 20 kHz, 30 kHz, 35 kHz or 40 kHz.
2. The high-frequency AC excites a piezoelectric stack inside the converter (also called transducer), which expands and contracts axially at the same frequency, converting the electrical signal into longitudinal mechanical motion.
3. The booster amplifies (or attenuates) the longitudinal amplitude of the converter through its mass-and-shape geometry, with no power loss in principle. Boosters are passive resonators.
4. The horn (sonotrode) couples the booster's amplified motion to the part. Like the booster, it is sized to be a half-wavelength resonator at the operating frequency, and it concentrates or distributes the amplitude over its working face. The face oscillates against the upper part and transmits the cyclic motion into the joint.

3.2 Heating mechanism

Heat at the joint comes from two superimposed sources. The first is intermolecular (viscoelastic) damping: every cycle of compression and shear strain inside the polymer dissipates a fraction of the input mechanical energy as heat, and this loss is sharply higher near the glass-transition temperature (T_g). The second is interfacial friction: the small asperities and the energy director at the joint surface rub against the mating part under pressure, and that surface friction dissipates additional energy as heat at exactly the location where it is wanted. The two mechanisms reinforce each other. A well-designed joint with a sharp energy director starts with high local pressure on a small contact area, surface friction dominates at the start, and as the tip of the director softens and spreads, viscoelastic heating in the molten film takes over until enough material has flowed and intermixed to form the bond.

3.3 Why amorphous resins behave differently from semi-crystalline resins

Amorphous polymers (ABS, PC, PMMA, PS, PSU, PEI) soften gradually over a wide range above their glass-transition temperature; they become rubbery, then leathery, then a viscous melt. This gradual softening is ideal for energy-director joints, because the tip of the director softens first, deforms, and then progressively transfers heat into a slightly wider zone as the parts collapse together. It also means the melt does not run away — it stays viscous enough to bond before it solidifies.

Semi-crystalline polymers (PE, PP, PA, POM, PET, PBT, PEEK, PPS) have a sharp, well-defined melting point because their crystalline phase melts at a discrete temperature. Below that temperature they are stiff and conduct vibration well; at and above it they go almost immediately to a low-viscosity melt that re-crystallizes quickly when the energy is removed. With an energy director, a semi-crystalline polymer's molten apex tends to cool and re-solidify before enough material has flowed across the joint, producing a brittle, weak interface. The accepted solution is the shear joint: the parts are designed with a small interference along a vertical wall, the horn drives the upper part axially, and the molten layer is continuously generated and smeared between two metal-cold surfaces that sit one above the other in a controlled telescoping motion. The continuous shear keeps melt available across the entire bond area until the controlled penetration depth is reached.

3.4 Frequency, amplitude and force — the three primary variables

At a given joint geometry, the rate of heat generation scales roughly with the product of frequency $\times$ amplitude squared. Force determines how much of that motion is converted into local strain at the joint instead of being lost as relative motion of the horn against the part. The selection of frequency and amplitude is governed by three trade-offs:

- Lower frequency (15–20 kHz) gives higher allowable amplitude (up to roughly 100–125 μm peak-to-peak), more energy per cycle, and the ability to weld larger parts and tougher materials. The trade-off is louder operation, larger and heavier stacks, and a higher risk of damaging delicate features.
- Higher frequency (30, 35, 40 or 70 kHz) gives smaller amplitude (typically 10–40 μm), gentler treatment of fine features, smaller and lighter stacks suitable for small parts, and quieter operation. The trade-off is less energy per cycle and a smaller maximum part size.
- Force is set so that the horn does not slip on the part and the energy director collapses in a controlled way. Too little force and the horn rattles on the part without coupling energy in; too much and the director collapses under static load before it can be melted, leading to cold welds.

3.5 The four phases of a weld cycle

An ultrasonic weld of an energy-director joint progresses through four characteristic phases, visible on the power-vs-time curve recorded by any modern generator:

5. Pre-trigger and contact: the horn descends, contacts the part, and builds up the trigger force before vibration starts. No melting yet.
6. Coupling and ramp-up: vibration starts, the energy director apex deforms, surface friction begins, and power rises sharply from the converter as the director starts to soften. This phase is short — typically 10–20 % of the total weld time.

7. Steady melt and flow: the director apex is fully molten, melt flows out from under the collapsing tip into the joint volume, the parts begin to collapse together, and generator power reaches a plateau. Most of the energy is delivered here.
8. Cool-down and hold: vibration stops at the chosen weld limit (time, energy or distance). The horn remains in place, holding force on the still-molten joint until it has solidified — the hold phase. Releasing force too early breaks the not-yet-solid weld; holding too long is harmless except for cycle time.

Rule of thumb: the four phases

Trigger – ramp-up – melt-and-collapse – hold. Watch the power trace; if it does not show a clean rise, plateau, and fall, something in the joint design or setup is wrong before you change any parameter.

4. Machine Construction and Working Principle

4.1 The five sub-systems

Every ultrasonic plastic welder, regardless of supplier or generation, consists of the same five sub-systems, arranged in a vertical press configuration. The fixture sits on the base; the part sits in the fixture; the horn is on the actuator above the part; and the actuator is mounted to a column or frame:

9. Generator (also called power supply): converts mains AC into high-frequency AC at the stack's resonant frequency. Modern generators are switching power supplies with integrated phase-locked-loop frequency control and digital control of amplitude, force, weld limit and cycle profile.
10. Stack: the resonant assembly of converter + booster + horn that delivers the vibrations to the part.
11. Actuator (press): the mechanism that lowers the stack, applies and controls force during the weld, and retracts at the end. Pneumatic-driven slides are still the most common; servo-driven slides are increasingly the standard for medical and other repeatability-critical applications.
12. Controller: the user interface and process logic, today almost always a colour touchscreen with a real-time display of power, force, distance and energy. It runs the weld in one or more control modes (time, energy, distance, peak power) and stores recipes.
13. Fixture (also called nest, anvil or holder): the lower tool that locates and supports the lower part during the weld. The fixture is application-specific and is usually designed and built per part.

4.2 The generator (power supply)

The generator's job is to drive the converter at exactly the converter's mechanical resonant frequency, even as that frequency drifts with stack temperature, horn loading and tooling wear. It does this with a phase-locked loop that continuously measures the phase between current and voltage at the converter terminals and adjusts the drive frequency to keep them aligned. The narrower the lock window, the more stable the weld. Output power ratings range from about 200 W for 40 kHz micro-welders to 6,000 W or higher for 15 kHz machines welding large parts.

Modern generators also implement digital amplitude control: instead of mechanically swapping boosters to change the amplitude, the user dials in a percentage on the controller (e.g., 80 %) and the generator scales the drive voltage to the converter so that the horn delivers 80 % of its maximum design amplitude. This makes amplitude an in-process tuning variable rather than a tooling decision.

4.3 The stack: converter, booster and horn

4.3.1 Converter (transducer)

The converter contains a stack of piezoceramic discs (almost always lead zirconate titanate, PZT) sandwiched and pre-stressed between a back mass and a front mass with a central tie-bolt. When the high-frequency drive voltage is applied across the discs, they expand and contract along the stack axis, producing axial motion at the front face. The converter's mechanical design is tuned so that the entire assembly is a half-wavelength resonator at the operating frequency. Typical converter output amplitudes are 15–25 μm peak-to-peak at 20 kHz.

4.3.2 Booster

The booster is a passive metal block (titanium, aluminium or steel) shaped as a half-wavelength resonator and bolted between the converter and the horn. Because the booster is necked down between its two ends, the velocity (and therefore amplitude) at the small end is greater than at the large end by the ratio of the cross-sectional areas. Boosters are specified by their gain ratio: 1:1 (no change), 1:1.5, 1:2 (most common), 1:2.5, sometimes up to 1:3. Reducing boosters (e.g., 0.6:1) are used to attenuate amplitude for delicate work. The booster also provides a node (a point of zero axial motion) where it can be clamped to the actuator without losing energy.

4.3.3 Horn (sonotrode)

The horn is the application-specific tool that contacts the part. It is also a half-wavelength resonator, machined from titanium, hardened steel or 7075-T6 aluminium. Its job is twofold: deliver the booster's amplitude into the part, and shape the contact face so that pressure and motion are uniform across the joint. Horn shapes include cylindrical, rectangular block ("bar horn"), stepped, exponential, catenoidal, spool, and complex profiled horns for specific parts. A finite-element analysis is performed for every new horn before it is machined to verify that the design's first longitudinal mode is at the target frequency (within typically ± 100 Hz at 20 kHz) and that there are no transverse modes nearby.

Material selection for the horn:

- Titanium (Ti-6Al-4V): the standard for high-amplitude, high-cycle, abrasive material applications. Excellent fatigue strength, good wear resistance, can run at the highest amplitudes.
- Aluminium (7075-T6): low cost, light, easy to machine. Used for prototype horns and large bar horns where mass is a concern. Wears quickly against abrasive fillers; can be carbide-coated for longer life.
- Hardened tool steel: used for staking, swaging and insertion tools where the tip wears against metal inserts; not generally used for full plastic welds because of its lower fatigue endurance at high amplitude.

Important property

The booster and horn are both passive resonators. They are tuned to a specific frequency and they do not store the energy — they pass it through. A horn that has been re-faced one too many times, or that has cracked, will detune; the generator will fight to lock and either fault out or weld inconsistently. Frequency check after every horn modification is mandatory.

4.4 The actuator (press)

The actuator is the mechanical column that lowers the stack onto the part, applies force during the weld, and retracts at the end. There are three architectures in common use:

- Pneumatic actuators are the workhorse. Force is set by a regulator on the supply air, and travel speed by a flow control. Cheap, robust and adequate for the great majority of welds, but they have limited control over the force profile during the weld, and the force will droop as the cylinder loses pressure during a fast collapse.
- Servo-pneumatic and pneumatic-with-position-feedback actuators add a linear encoder to a pneumatic cylinder so that distance-based control becomes available, while keeping the simplicity of compressed air. Many medical-grade welders use this architecture.
- All-electric servo actuators replace the cylinder with a ball-screw and brushless servo motor. They give programmable trigger force, programmable force during the weld (force ramps, dual-force profiles), programmable trigger distance, very repeatable down-speed, and absolute closed-loop control of weld distance to a few microns. Servo actuators are now standard for medical, micro-fluidic and high-value automotive parts where every weld must be data-logged and traceable.

4.5 The controller and control modes

The controller is the brain that decides when each weld phase ends. The choice of control mode — what the controller is monitoring to decide "weld is finished" — is one of the two or three most important decisions in process design, because it determines what the welder is actually holding constant from cycle to cycle:

Mode	Weld terminates when...	Holds constant	Best for	Watch out for
Time	A pre-set time elapses	Energy delivery time	Stable cosmetic welds, fixed-cycle production	Compensates poorly for material lot variation; weld depth drifts with horn wear
Energy	Cumulative joules reach a target	Total energy into the joint	Most general-purpose welds; very common default	If the trigger force or amplitude varies, the energy needed for a good weld will too
Distance / collapse	The horn travels a programmed	Final part dimension and weld depth	Hermetic seals, medical, where	Requires a stiff press and a good encoder; needs a servo or

Mode	Weld terminates when...	Holds constant	Best for	Watch out for
	distance into the part		final geometry is the spec	position-feedback actuator
Peak power	Generator output reaches a peak then drops by a programmed percent	Indirectly, the moment of full melt	Diagnostic and tuning runs	Rarely used as a primary production mode
Ground detect	Horn touches the lower fixture (last 0.1–0.2 mm)	Final closure	Films, foils, very thin parts	Risks horn-to-fixture damage if mis-set

Most-recommended starting point for new applications

Set the welder to ENERGY mode for joint development and process characterization (it is the most forgiving and the most informative). Once the process is stable and the part design is finalized, switch high-runner production to DISTANCE / COLLAPSE mode if the actuator supports it — distance control gives you a final part dimension that is independent of small lot-to-lot variation in polymer, ambient temperature and horn wear.

4.6 The fixture (nest, anvil, holder)

The fixture has four jobs, all of them critical: locate the lower part repeatably, support the joint area rigidly so that the ultrasonic energy is transmitted into the joint and not absorbed by deflection, prevent damage to delicate cosmetic surfaces, and reflect ultrasonic energy back into the joint rather than absorbing it. A common rookie mistake is to build the fixture from soft, machinable plastic for cheapness and speed; soft fixtures flex, absorb energy and produce cold welds. Fixtures should be machined from aluminium tooling plate or steel, with conformal pockets or replaceable urethane inserts only at the cosmetic-contact surfaces. The fixture must also clear the path of the horn during collapse — if the horn bottoms out on the fixture, the booster, horn or both will be damaged.

Fixture rule of thumb

If you can deflect the fixture by hand under the joint area, the fixture is too soft for ultrasonic welding. The shop maxim is: "The fixture must be at least an order of magnitude stiffer than the joint it is supporting."

4.7 Frequency selection

The choice of operating frequency is a tooling-level decision (it determines the entire stack and most of the actuator) and is rarely changed once a production line is built. Selection is driven by part size, feature delicacy and material:

Frequency	Typical max amplitude (μm pk-pk)	Typical part size	Best applications	Notes
15 kHz	100–125	Up to ~250 mm (10 in)	Large automotive parts, instrument panels, soft polyolefins, large packaging	Loud; requires sound enclosure; heavy stack
20 kHz	60–100	Up to ~150 mm (6 in)	Workhorse for general industry, automotive, white goods, medium toys, packaging seals	By far the most common — over 60% of installed base
30 kHz	30–60	Up to ~75 mm (3 in)	Smaller medical disposables, electronics housings, finer-feature consumer goods	Quieter than 20 kHz; better for fine cosmetic features
35 kHz / 40 kHz	15–40	Up to ~50 mm (2 in)	Micro-fluidics, small medical, sensor housings, button assemblies	Below human hearing threshold of harmful sound; minimal enclosure needed
70 kHz	5–15	Up to ~25 mm (1 in)	Micro-electronics, semiconductor packaging, micro-fluidic devices	Specialized; few suppliers; high cost per machine

5. Materials and Compatibility

5.1 What can and cannot be welded

Only thermoplastics can be ultrasonically welded. Thermosets, vulcanized rubbers and most elastomers cannot be welded because their cross-linked molecular networks do not soften reversibly with heat — instead they char, degrade or crumble. Within the thermoplastic family, weldability ranges from excellent to marginal, and selection depends on whether the polymer is amorphous or semi-crystalline, what fillers and additives it contains, and how much moisture it has absorbed.

5.2 Amorphous thermoplastics (the easy ones)

Amorphous thermoplastics have no crystalline order; they soften over a wide range of temperatures starting at the glass-transition temperature (T_g) and produce a viscous melt that flows controllably. They couple ultrasonic vibration efficiently because they are stiff and lossy in the right proportion. Energy-director joints work very well with them. The most weldable amorphous polymers are:

Polymer	Common name	T_g / softening (°C)	Weldability with itself	Notes
ABS	ABS	~105	Excellent	Industry default; welds at low to medium amplitude
PC	Polycarbonate	~150	Excellent	Sensitive to moisture; pre-dry if humid
PMMA	Acrylic	~105	Excellent	Brittle melt; needs a controlled hold
PS-GP / PS-HI	Polystyrene	~100	Excellent	The original ultrasonic-friendly material
SAN	Styrene-acrylonitrile	~105	Excellent	Welds like ABS
PSU / PES	Polysulfone / polyethersulfone	~190 / 220	Very good	Higher amplitudes needed; stable in autoclave
PEI	Ultem (polyetherimide)	~217	Very good	Demanding amplitude, but joints are very strong
PPO / PPE blends	Modified PPE	~150	Excellent	Often used in white goods; very stable
PVC (rigid)	uPVC	~80	Good with care	Watch overheating; can release HCl

5.3 Semi-crystalline thermoplastics (the demanding ones)

Semi-crystalline thermoplastics have a defined melting point (T_m) and a low-viscosity melt that re-crystallizes quickly. They are hard to weld with energy-director joints because the molten apex re-solidifies before enough material has flowed across the joint. They almost always need shear joints and they need higher amplitude (usually doubled compared with the equivalent amorphous polymer). They also need very tight tolerance on the shear-joint interference because the melt zone is thin.

Polymer	Common name	T_m (°C)	Weldability with itself	Recommended joint
PE-LD / PE-HD	Polyethylene	115 / 135	Good (LD), fair (HD)	Shear joint; high amplitude
PP	Polypropylene	165	Good	Shear joint; high amplitude
PA6 / PA66	Nylon 6 / 66	220 / 260	Fair to good	Shear; pre-dry to <0.2 % moisture or weld will foam
POM	Acetal / Delrin	175	Fair	Shear; degradation gas can cause voids
PET / PBT	Polyester	255 / 225	Fair	Shear; brittle joints if overheated
PEEK	Polyetheretherketone	343	Good	Specialized; high amplitude, high force
PPS	Polyphenylene sulfide	285	Fair	Hard to weld; shear joint and high force
PTFE	Teflon	327	Very poor	Surface treatment required; usually not weldable in production

5.4 Welding dissimilar materials

Two parts of different polymers can be welded only if their melt temperatures are within a small window (typically within 20 °C for amorphous-to-amorphous, and they must be chemically compatible at the molecular level — they need to mix on a polymer-chain scale, not just lie side-by-side). The well-established compatible pairs are:

- ABS to PMMA — well-known, frequently used for transparent lenses on coloured housings.
- ABS to PC — common for housings; needs ABS amplitude settings.
- ABS to SAN — essentially identical chemistry; welds easily.
- PC to PMMA — possible but marginal; tightly controlled.
- PS to SAN — possible; common in medical disposables.

Polypropylene and polyethylene look chemically similar but do not weld together because their melts are not miscible at the polymer-chain level. PA cannot be welded to anything but itself. Filled and unfilled grades of the "same" polymer are also dissimilar in welding terms — they will weld, but the joint strength will be lower than the unfilled-to-unfilled equivalent.

5.5 Effect of fillers, additives and reinforcements

- Glass fibre: up to 10–15 % is fine, 20 % is workable, 30 % is the practical upper limit. Above this the fibres absorb and scatter ultrasonic energy and the joint becomes a fibre-rich resin-poor seam.
- Mineral filler (talc, calcium carbonate): similar limits; talc tends to be more forgiving than glass.
- Flame retardants: most flame retardants reduce weld strength by 20–40 % because they raise melt viscosity and char before flowing. Brominated systems are worst; phosphate systems are typically less harmful.
- Pigments: titanium dioxide (white) and most colourants are fine. Carbon black at high loading reduces weld strength and can cause spot heating.
- Internal lubricants and mould-release: even small amounts of slip agents at the weld surface contaminate the joint. Trial parts should be welded as molded — never wiped with mould-release.
- Regrind: every regrind cycle drops molecular weight and weldability. Most resin suppliers and the AWS guidance recommend no more than 25 % regrind in critical welded parts.

5.6 Moisture and the special case of hygroscopic resins

Polyamides (PA6, PA66) and polycarbonate are hygroscopic — they absorb water from ambient air. Even 0.2–0.3 % moisture will steam-blow the joint during welding, producing a foamy, weak weld surface. Polyamide that has been sitting on the shop floor for a week may have absorbed 1–2 % moisture, which is enough to fail every weld. The fix is to weld within 24 hours of molding (the ideal), or to dry parts before welding (lower-temperature, longer-time, e.g. 80 °C for 8 hours for PA6, with sealed bags between drying and welding). PEEK and PSU are mildly hygroscopic; the rule of thumb is to dry any nylon or polyamide if the parts are more than a day old.

Quick material check before you start welding

1) Same polymer on both sides? 2) Amorphous or semi-crystalline (decides energy director vs. shear)? 3) Filler less than 20%? 4) Flame retardants and additives at minimum? 5) Hygroscopic and dried/recent? — answer those five questions before touching the press.

6. Joint Design — Geometries that Define the Weld

The joint is the small region of the part where the two halves meet, and it is the single most important design feature for ultrasonic welding. A correctly designed joint focuses the ultrasonic energy onto a small initial contact, melts in a controlled and repeatable way, contains the molten polymer where it should flow, and finishes with a defined geometric stop. Four joint families are used in practice — butt with energy director, step with energy director, tongue and groove, and shear. Within each family there are variants for hermetic seals, for cosmetic finish, for film welding and for self-locating geometries. Picking the right joint is a function of polymer (amorphous vs. semi-crystalline), strength requirement, sealing requirement, cosmetic requirement and the volume of the part.

6.1 The energy director — the universal feature

An energy director is a small, sharp triangular ridge molded into one of the two mating surfaces. Its function is to concentrate the ultrasonic energy onto a tiny initial contact area at the apex of the triangle. A small initial contact under high local pressure means high local stress, intense surface friction at the start of the cycle, rapid local heating, and a controlled propagation of melt outwards from the apex into the rest of the joint. Without an energy director, an ultrasonic weld between two flat surfaces will fail — the energy will be spread over too large an area for any one location to reach melt temperature in a reasonable time, and the parts will simply rattle against each other while the bulk of the energy is dissipated as audible noise.

6.1.1 Standard dimensions

The geometry of the energy director is calibrated to the polymer family. The two parameters are the included angle at the apex and the height of the director. Industry-standard values, as published in the AWS G1.1M guide and the design literature of every major OEM (Branson/Emerson, Dukane, Herrmann, Sonics & Materials, Rinco, Telsonic), are:

Parameter	Amorphous resins	Semi-crystalline resins (when energy director used at all)	Notes
Included angle	90°	60°	60° concentrates energy more sharply, helps the lower-viscosity melt
Height (h)	0.40 – 0.60 mm typical (function of part size)	0.50 – 0.80 mm	Larger parts → larger directors; rule of thumb: $h \approx 0.05 \times \text{wall thickness}$
Base width	$= 2 \times h \times \tan(\text{angle}/2)$	$= 2 \times h \times \tan(\text{angle}/2)$	Falls out of geometry
Apex radius	≤ 0.05 mm (sharp)	≤ 0.05 mm (sharp)	Sharper apex = lower energy needed; mold

Parameter	Amorphous resins	Semi-crystalline resins (when energy director used at all)	Notes
			maintenance is critical here
Min. distance from edge	$\geq 1.0 \times$ wall thickness	$\geq 1.0 \times$ wall thickness	Prevents flash blow-out

The energy director must be molded onto only one of the two parts. The mating surface is a flat (or for a tongue-and-groove, a flat-bottomed groove). Putting energy directors on both halves leads to an unstable, asymmetric melt and typically a weak joint.

6.1.2 Variants

- Continuous director: a single ridge running around the perimeter — the standard for hermetic seals because it produces a continuous melt line.
- Interrupted director: short segments separated by gaps; reduces the energy needed and is appropriate where strength is needed but a hermetic seal is not.
- Cross-hatch / criss-cross directors: two perpendicular sets of small directors used for film-to-rigid welding and for some textile applications; sacrifice cosmetic finish and sealing for weld strength on thin films.
- Multiple parallel directors: two or more parallel ridges; used on wide, thin walls to ensure that the joint does not collapse along just one ridge. Each director is dimensioned at about 60% of the equivalent single-director size.

6.2 Butt joint with energy director

The butt joint with energy director is the simplest and most common ultrasonic weld geometry. The lower part presents a flat upper surface; the upper part presents a triangular ridge that touches the lower part along a thin line. When ultrasonic energy is applied, the apex of the triangle melts, the molten polymer flows outwards, the upper part collapses by approximately the height of the director, and the two parts fuse. The collapse is typically 0.4–0.6 mm — equal to the director's height — and is the visible signature of a good butt-joint weld.

Strengths and limitations of the butt joint:

- Very simple to mould: only one extra feature (the triangular ridge) on one of the two parts.
- Tolerant of moderate misalignment between parts during assembly.
- Excellent for amorphous resins; usable for semi-crystalline only with restrictions.
- Joint strength: typically 60–80% of the parent material's tensile strength when correctly executed.

- Cosmetic limitation: the molten flash flows outward to both sides of the joint and is visible unless hidden by a step.
- Sealing limitation: a butt joint with a continuous director is hermetic if welded properly, but the seal is small and any contamination ruins it.

6.3 Step joint with energy director

A step joint is a butt joint where one part has a small lip that overlaps the other. The energy director is on the horizontal face of the joint, but a vertical step on the outside of the joint hides any flash that flows outwards, gives self-location of the upper part on the lower part during loading, and significantly improves the cosmetic appearance. The step depth is typically 0.5–0.8 mm and the step clearance (horizontal slip-fit) is 0.05–0.10 mm so that the parts go together easily but do not rattle.

Strengths and limitations of the step joint:

- Better cosmetics: flash is hidden behind the step.
- Self-locating: the step centers the parts on the press, which improves repeatability.
- Stronger than butt: the step itself contributes to shear strength on top of the welded interface.
- Slightly more demanding mould tolerances on the step than on a plain butt.
- This is the joint experienced welding engineers (e.g., Sonics & Materials, per the published Assembly Magazine 2019 round-table) most often recommend as the default for new amorphous-resin parts.

6.4 Tongue-and-groove joint

The tongue-and-groove joint has a small tongue molded onto one part that fits into a corresponding groove on the other. The energy director sits on the end of the tongue or on the bottom of the groove. The geometry hides flash on both sides of the joint (internal and external), so it gives the cleanest cosmetic appearance of any of the standard joints, and it gives excellent self-location. It is the default choice for high-end consumer products and for parts where flash is unacceptable on either side.

Strengths and limitations of the tongue-and-groove joint:

- Best flash containment: no visible flash inside or outside.
- Strongest joint of the four standard types when correctly designed.
- Good self-location for assembly.
- Tightest mould tolerances of the four — the tongue width and groove width must match within 0.05 mm to avoid binding or excess clearance.
- Most expensive to mould (more cores, more parting-line decisions).

6.5 Shear joint — the answer for semi-crystalline polymers

The shear joint is the only practical answer for ultrasonic welding of semi-crystalline polymers (PE, PP, PA, POM, PET, PBT). Instead of an energy director on a horizontal face, the shear joint has a small interference between two vertical walls — typically 0.2–0.4 mm of total interference, distributed as 0.1–0.2 mm per side. As the upper part is driven down by the horn, the small interference is continuously melted and the molten polymer is smeared along the vertical wall as the parts telescope together. The continuous shearing keeps fresh melt available across the whole joint surface for the duration of the weld, which prevents the molten film from re-solidifying before bonding (the failure mode that kills energy-director joints on semi-crystalline resins).

Key design points for shear joints:

- Total interference: 0.2–0.4 mm. Less and the joint is weak; more and the part stalls before reaching weld depth.
- Lead-in chamfer or radius: $0.5 \text{ mm} \times 30^\circ$ at the entry to the joint, to allow self-location and to start the shearing motion progressively.
- Weld depth (telescoping distance): $1.25 \text{ to } 1.5 \times \text{wall thickness}$ for a strong joint; greater depth for hermetic seals.
- Outside support: the lower part's outside wall must be supported by the fixture in the joint area, otherwise the lower wall flexes outward and the interference is lost.
- Wall thickness uniformity: 0.05 mm tolerance on the joint walls is required for repeatable strength. This is more demanding than it sounds and frequently dictates the molding process.
- Cycle time: shear joints take 3–4× longer than butt joints because more polymer must be melted and displaced. A butt joint that welds in 0.3 s might take 1.0–1.2 s as a shear joint.

Quick joint-selection guide

Amorphous polymer + general use → Step joint with energy director. Amorphous polymer + clean cosmetics → Tongue and groove. Amorphous polymer + simple geometry → Butt with energy director. Semi-crystalline polymer (PP, PE, PA, POM, PET) → Shear joint, no exceptions for production parts.

6.6 Hermetic seal design

A hermetic ultrasonic weld must produce an unbroken molten line all the way around the part. The two requirements are continuity of the energy director (no gaps anywhere) and uniform coupling of ultrasonic energy along the entire perimeter. Practical rules:

- Use a continuous director with no breaks.

- Joint perimeter geometry should have radii at every internal corner ($R \geq 1.0$ mm); sharp corners create energy concentration and burn-through.
- Wall thickness around the perimeter should be uniform within $\pm 10\%$ — varying wall thickness leads to varying energy coupling and a non-uniform melt line.
- Plan a leak test (gross-leak air-pressure decay or helium for medical) into the inspection regime, not as an afterthought. A nominal hermetic design cannot be assumed hermetic without testing.

6.7 Joint design for staking, swaging, insertion and spot welding

Ultrasonic energy can also be used in four assembly operations that are not strictly welding but use the same equipment:

- **Staking:** a plastic boss is melted and reshaped over a sheet-metal or PCB hole to retain it. The horn tip is profiled (rosette, dome, hollow, knurled, flush, low-profile) to give the desired finished head.
- **Swaging:** a continuous plastic ridge is folded over to capture another component, e.g., a lens in a frame.
- **Insertion:** a metal threaded insert is pressed into a slightly undersized molded hole while ultrasonic energy melts the surrounding plastic; the polymer flows into the insert's knurls and locks it in place after solidification.
- **Spot welding:** a single small horn tip is plunged into the upper part; the heat melts both parts at the contact point and a small spot weld is formed. Used to join two flat sheets without a molded joint.

7. Design for Manufacturing (DFM) Guidelines

Design for Manufacturing for ultrasonic welding is the discipline of building features into the part — at the CAD stage, before any tooling is committed — that make the weld robust, repeatable and inexpensive. The dominant authorities here are the design guides published by Branson/Emerson (the de facto industry reference), Dukane, Herrmann Ultrasonics and Sonics & Materials, and the AWS G1.1M guide which consolidates much of that material. The following rules summarize their consensus.

7.1 Wall and joint geometry rules

- Joint location: place the joint as close as possible to the horn contact face. Every millimetre of wall the energy must travel through is energy lost to damping. The longest reasonable distance for a near-field weld is roughly 6 mm for amorphous resins and 3 mm for semi-crystalline; beyond that, the weld becomes a far-field weld and the rules change.
- Wall thickness: aim for 1.5–3.0 mm at the joint. Thinner walls cannot transmit enough energy without flexing and flashing; thicker walls absorb energy and waste cycle time.
- Wall thickness uniformity: $\pm 10\%$ within the joint perimeter. Variation drives varying weld depth and varying strength.
- Generous radii on internal corners (≥ 0.5 mm minimum, 1.0 mm preferred). Sharp internal corners are stress concentrators that fail under cyclic ultrasonic load.
- Ribs and bosses near the joint: keep them at least one wall thickness away. Ribs and bosses transmit and reflect ultrasonic energy unpredictably and can cause local hot spots, melting or cracking.
- Avoid sharp transitions in cross-section near the joint. A 3 mm wall stepping down to a 1 mm wall just above the joint will flex during welding; use a tapered or fillet-blended transition.

7.2 Flash control

Flash is the molten polymer that escapes the joint volume. It is unavoidable to some degree, but it should be predictable and contained. The four common flash-control features are:

- Flash trap: a small recess machined or molded into the lower part next to the joint, below the joint surface, into which excess melt can flow. The recess is open downward and so is invisible from outside.
- Step around the joint (built into the step joint, see §6.3): hides flash on the outside.
- Tongue-and-groove geometry (§6.4): hides flash on both sides.
- Reduced energy director size for cosmetic parts: a smaller director produces less flash, but also less molten material, so weld strength drops. Trade-off.

7.3 Gating, parting line and venting

- Place the gate as far away from the joint as possible. The flow lines and orientation near the gate are different from elsewhere; if the gate is at the joint, the polymer at the joint will have non-uniform mechanical properties.
- Keep the parting line clear of the energy director. A flash line on the apex of the energy director destroys its sharpness and the weld will be inconsistent.
- Vent the joint area in the mould: trapped gas creates short-shots and incomplete energy directors, which present as random weak welds.
- Avoid weld lines (knit lines from the moulding process) crossing the joint perimeter — they are mechanically weak and they fail first under ultrasonic load.

7.4 Material selection and grade callouts

- On the drawing, specify the polymer family, grade, melt-flow index window, regrind percentage limit (typ. $\leq 25\%$), and any drying requirement.
- If the polymer is hygroscopic (PA, PC), specify the maximum allowable moisture at welding (typ. $\leq 0.20\%$) and the time-to-weld after moulding (typ. < 24 h) or the drying procedure if longer.
- Where a flame retardant is required, specify a non-halogenated grade if at all possible — they typically reduce weldability less than brominated grades.
- Filler content above 20 % requires consultation with the welder and material suppliers; redesign the joint with a wider energy director and accept lower strength.

7.5 Coupling and energy delivery

- Provide a flat horn-contact area on the upper part directly above the joint. The horn-contact area should be the same width as the joint or wider, and parallel to the joint plane within $\pm 0.1^\circ$ (parallelism error becomes amplitude variation across the joint).
- Avoid placing the horn on a thin or unsupported wall. The wall under the horn must not flex while energy is being applied.
- Avoid free-standing tall walls or rib structures within $\frac{1}{4}$ wavelength of the horn contact. They will resonate sympathetically and crack.
- Keep cores and wall sections symmetric in the joint area — asymmetry produces bending modes in the joint instead of the desired axial compression.

7.6 Other geometric DFM rules

- Draft angles on the joint walls: $\geq 0.5^\circ$ draft on the outside of the joint to ease ejection from the mould without disturbing the joint geometry.
- Datum scheme: the part's primary datum should be the horn-contact face or a reference parallel to it. The fixture references the lower part to the same datum; this maintains parallelism through the press.

- Avoid metal inserts in the joint zone. Metal at the joint reflects ultrasonic energy and can damage the horn over time.
- Plan for a 0.5–1.0 mm overall "weld collapse" in the assembled-part dimensions. The two halves before welding must sit slightly proud of the final assembled height by exactly the planned collapse.

Top-six DFM checklist before tooling release

1) Joint type matched to polymer family. 2) Energy director (or shear interference) sized correctly. 3) Wall thickness uniform within $\pm 10\%$ at the joint. 4) Horn contact face flat, parallel and supported. 5) Flash control feature included (step, tongue-and-groove, or trap). 6) Gate and weld lines moved away from the joint.

8. Tolerances and Dimensional Considerations

Ultrasonic welding is more sensitive to part-to-part dimensional variation than most assembly processes, because the energy delivered by the horn is fixed in time, energy or distance — variations in the joint geometry directly translate into variations in weld outcome. The tolerance budget below is the consensus from the published OEM design guides; tighter tolerances may be required for hermetic, medical or critical-strength applications.

Dimension	Recommended tolerance	Why this matters
Energy director height	± 0.05 mm	Director height directly sets weld energy and collapse distance
Energy director apex angle	$\pm 5^\circ$	Apex angle changes the contact area at start of weld
Apex sharpness (radius)	≤ 0.05 mm	A rounded apex spreads contact and lengthens cycle
Joint flatness across one part	≤ 0.1 mm over 100 mm	Non-flat joints contact in patches; uneven melt
Parallelism between mating surfaces	≤ 0.10 mm or 0.1° , whichever is tighter	Non-parallel joints concentrate energy at one side
Wall thickness in joint zone	$\pm 10\%$ (typically ± 0.10 mm)	Drives local stiffness and energy coupling
Step / shear lead-in clearance	+ 0.05 / -0.00 mm	Press fit in the lead-in is the failure mode
Shear-joint interference (per side)	0.1–0.2 mm + 0.05 / -0.00	Too much = stalled weld; too little = no weld
Tongue-and-groove fit	+ 0.05 / -0.00 mm	Tighter than step; controls flash
Pre-weld assembly height (above target)	± 0.10 mm	Drives the collapse distance the press will see
Mating surface roughness	Ra ≤ 1.6 μ m on flat	Higher roughness needs higher amplitude

8.1 Tolerance stack-up considerations

Two parts with individual tolerances of ± 0.05 mm on energy director height and ± 0.10 mm on wall thickness can stack up to ± 0.20 mm of variation at the assembly. With the welder set in time mode this will produce visibly different welds. Tolerance stack analysis should always include energy-director height + parallelism + lead-in clearance, and the result should fit within the dynamic range of the chosen control mode. Distance / collapse mode is the most forgiving

of stack-up because it compensates by always reaching the same final geometry; time mode is the least forgiving.

8.2 Mould design and validation tolerances

The energy director feature in the mould wears under production. New moulds reproduce the apex sharpness perfectly; after 100,000–500,000 shots the apex starts to round. A common production failure mode in ultrasonic welding is "the welds got worse over the year" — and the answer is that the energy director is no longer 0.5 mm tall with a sharp apex but 0.4 mm tall with a 0.10 mm radius, because the mould's matching V-groove has worn. Mould validation should include CMM measurement of the energy director's height, angle and apex radius at PPAP, and again at every preventive-maintenance interval (typ. every 250,000 shots). Replaceable inserts at the energy director are a good design practice for high-volume parts.

9. Process Parameters and Tuning

After joint geometry, the second-most-important determinant of weld quality is the choice of process parameters. Modern generators expose seven to ten parameters per recipe; in practice three of them (amplitude, force, weld limit) carry most of the weight, and the remaining ones (trigger force, ramp profile, hold time, hold force) are fine-tuning. The procedure below is the standard development sequence used across the industry.

9.1 The seven primary parameters

Parameter	Typical range	What it controls	How to tune
Amplitude	20–80 μm pk-pk at 20 kHz; 20–40% of max for delicate work	Rate of melt at the joint	Start at supplier-recommended value for the polymer; raise until the weld is consistent and just below burn-through
Trigger force	20–500 N (small parts) up to 3000+ N (large parts)	When ultrasonics start	Just enough to seat the parts and prevent slip — too high crushes the energy director before it can melt
Weld force	Same as or slightly above trigger	Coupling during the weld; rate of collapse	Tune in concert with amplitude; high force + low amplitude = crisp welds; low force + high amplitude = soft welds
Weld limit (time / energy / distance)	0.1–1.5 s / 5–500 J / 0.3–1.5 mm	When ultrasonics stop	Choose the mode first (see §4.5); tune the value by welding parts and measuring strength or seal
Hold time	0.2–1.0 s	Cooling under pressure	Long enough for the joint to solidify; longer is harmless except cycle time
Hold force	Same as weld force, sometimes higher	Joint compression during cooling	Often left equal to weld force; raise it slightly for hermetic seals
Pre-trigger / ramp	0–50 ms	Soft start of ultrasonics	Used on delicate parts to prevent shock; usually 0 for general work

9.2 Standard development sequence

The standard sequence to develop a new ultrasonic weld recipe (AWS G1.1M section 8 covers the formal version of this; OEM training courses teach the same logic) is:

14. Pick the control mode based on requirements: ENERGY for general development; DISTANCE for hermetic, medical or critical dimensional control; TIME only for very stable, well-known production.
15. Set amplitude at the polymer-family default (e.g., 50% for ABS, 80% for PP, 100% for PA).
16. Set trigger and weld force at a moderate value (200–400 N for typical small parts).
17. Run a parameter sweep on weld limit: weld 5–10 parts at five different time / energy / distance values bracketing the expected setpoint, pull-test or leak-test each. The strength-vs-weld-limit curve typically rises, plateaus, then falls (over-welding causes burn-through).
18. Pick the weld limit at the start of the plateau — the point of maximum strength with minimum cycle time and material flow.
19. Repeat the sweep for amplitude with weld limit fixed at the chosen value; pick the amplitude that gives the tightest joint-strength distribution (lowest standard deviation) at that plateau strength.
20. Repeat for force; the force optimum is usually broad. Pick a value at the top of the broad plateau.
21. Tune hold time to be just long enough for the joint to solidify (verified by pulling on the joint immediately after the cycle — if you can deflect the joint, hold longer).
22. Lock the recipe; run a 30-piece capability study; calculate Cpk on the strength or seal measurement. Target Cpk ≥ 1.33 for industrial, ≥ 1.67 for medical.

9.3 Tuning the stack — a separate, periodic exercise

Independently of recipe tuning, the stack itself must be tuned and rechecked periodically. "Tuning" the stack means measuring its mechanical resonant frequency (with a sweep / impedance analyzer built into modern generators) and confirming it is within the lock window of the generator. A new stack tunes once at first use, after every horn change, after every booster change, and any time the welder reports lock errors or chatter at the start of the cycle. Modern generators (Branson DCX, Dukane iQ, Herrmann HiQ, Telsonic MAG) auto-tune at startup and also run a rapid auto-tune at every cycle start, but the operator should still verify the result on a calibrated scope or amplitude-meter at scheduled intervals (e.g. weekly).

Distinguish recipe-tuning from stack-tuning

Recipe-tuning sets amplitude, force and weld limit for the part. Stack-tuning sets the converter / booster / horn at their resonant frequency. The two are independent: changing the recipe does not retune the stack, and retuning the stack does not change the recipe.

9.4 Reading the power-vs-time trace

Every modern generator records and displays a power-vs-time trace for each weld. Reading this trace is the fastest diagnostic available, and it is the first thing an experienced ultrasonic engineer looks at when a process drifts:

- Initial spike too low → energy director was crushed before the weld started (force too high, or director collapsed during loading).
- Initial spike too high, no plateau → not coupling, parts rattling (force too low, or contact face misaligned).
- Smooth rise, no plateau, runs to time-out → polymer is the wrong grade, or amplitude is too low to reach melt.
- Plateau with second peak at the end → over-welding; flash is bridging the lower part to the fixture and is being re-energized.
- Erratic trace, cycle to cycle → a mechanical fault in the stack, the actuator or the fixture; parts are not being presented consistently.

10. Setup, Fixtures and Tooling

10.1 Initial setup of the welder

Setup of an ultrasonic welder for a new part is a structured procedure. The published sequence (AWS G1.1M Annex; Branson, Dukane and Herrmann training material is consistent with each other on this) is essentially:

23. Mechanical: install the horn, booster and converter as a single tested stack. Verify torque on the stud joint between converter-booster (typ. 25 N·m) and booster-horn (typ. 30 N·m). Use a torque wrench, never "feel".
24. Electrical: connect the converter cable; verify the generator type matches the converter.
25. Auto-tune: run the generator's auto-tune routine. Resonant frequency should fall within the rated window (e.g., 19,800–20,200 Hz for a nominal 20 kHz stack).
26. Amplitude check: ideally with an amplitude meter (calibrated optical pickup), verify the horn face amplitude at 100% setting matches the design value within $\pm 5\%$.
27. Mechanical alignment: bring the horn down to within 1 mm of the fixture; verify visual parallelism. Bring it down further onto a flat shim of soft aluminium or paper, drop a few light strikes, and inspect the impression for uniform contact.
28. Fixture install: bolt down the part fixture; verify the joint plane is level.
29. Place the parts; lower the press; verify the horn touches only the intended contact face on the upper part with no other interference.
30. Run a few welds with the recipe at a conservative low-energy setting; iterate up to the chosen recipe.

10.2 Horn design considerations

Horn design is a specialty. Most plant engineers do not design horns; they specify the contact geometry to the welder OEM or a horn-design house, who run a finite-element analysis and machine the horn. The plant engineer should still understand:

- Gain: the horn itself can amplify amplitude, by the same area-ratio principle as the booster. Stepped and exponential horns have gains up to about 5; complex contoured horns rarely exceed 2.
- Slot patterns: large-area horns are often slotted to suppress transverse modes that would crack the horn at high amplitude. The slot pattern is part of the modal design.
- Contact face: replaceable tips for staking and insertion; integral profile for plastic welding. The face is normally surface-hardened or carbide-tipped for abrasive materials (glass-filled resins).
- Stud joint to booster: the most-stressed joint in the entire stack. A loose stud is the most common cause of horn failure (cracking from the threaded stud hole). Re-torque on every horn change.

10.3 Booster selection

Boosters provide three discrete amplitude scaling steps: typically 1:1.5, 1:2, and 1:2.5. The choice is driven by the horn's own gain and the application's required amplitude. Industry practice is to pick the smallest booster gain that, combined with the horn gain and the generator amplitude setting, gets the target horn-face amplitude with the generator running between 50% and 90% of its amplitude range. Running the generator at 100% all the time means there is no headroom for material variation; running below 30% means the booster is too small and amplitude-control resolution is poor.

10.4 Fixtures (nests)

Fixtures locate the lower part, support it during the weld, and protect cosmetic surfaces. Best practice rules:

- Material: machined aluminium tooling plate (6061-T6) or steel for production fixtures. Avoid plastics, urethane bulk, or compliant materials in the load path.
- Locate the part on at least three datum features; clamp positively but lightly (a clamp that bears 100 N is enough for most small parts; over-clamping distorts thin walls).
- Pocket the fixture so that the joint area is fully supported from below — the support area should match the joint footprint within 1 mm. Unsupported joints flex and waste energy.
- Provide an air-blow-off port to clear flash and dust between cycles for high-volume production.
- Provide a vertical clearance for horn over-travel: if the horn ever bottoms out on the fixture, the impact damages the horn. Build a 0.5–1.0 mm gap below the lowest expected horn position.
- On the fixture drawing, specify the location of cosmetic-protect inserts (urethane, soft polymer or felt) only at the cosmetic-contact areas. They must not be in the load path of the joint.

10.5 Operational environment

Ultrasonic welders are robust but they perform best in stable conditions:

- Compressed air supply: clean, dry, regulated. 6–7 bar typical; 0.5 μm particulate filter; oil-free or oil-free-to-class-1 air. Variable pressure leads to variable force, leads to variable welds.
- Power: stable mains, ideally on its own breaker. Generators are switching power supplies and tolerate dips, but heavy adjacent loads (resistance welders, capacitor banks) will affect lock stability.
- Temperature: 18–28 $^{\circ}\text{C}$; the resonant frequency of titanium horns shifts at roughly 2 $\text{Hz}/^{\circ}\text{C}$ — not enough to matter in a controlled factory, but a horn taken from a 5 $^{\circ}\text{C}$ warehouse to a 30 $^{\circ}\text{C}$ welder will need to re-tune.

- Cleanliness: dust on the joint surface, mould-release residue, and oil from operators' hands all reduce weld quality. Establish glove and handling protocols.
- Sound: 20 kHz operation is at the edge of human hearing; harmonics at 40 kHz are not audible but not harmless. ACGIH limits occupational exposure to 110 dB at 20 kHz for 8 hours. Sound-attenuating enclosures are standard for 20 kHz, optional for 30–40 kHz.

10.6 Operator training and safety

Pinch points (the horn closing onto the fixture) and high-frequency sound are the two safety concerns. Modern welders have two-hand anti-tie-down controls, light curtains or interlocked enclosures by default. The operator's role is loading and unloading; everything else (force, amplitude, time) is set by the engineer in a password-protected section of the controller. Modern controllers allow per-recipe lock-out so that the operator cannot inadvertently change a parameter.

11. Quality Inspection and Testing

Quality control of ultrasonic welds combines in-process monitoring (using the data the welder itself records) with destructive and non-destructive tests on sample parts. The published guidance is in AWS G1.1M (general inspection guidelines), AWS B2.4 (qualification of welding procedures and operators for thermoplastics), and the relevant industry-specific standards (ISO 13485 for medical, IATF 16949 for automotive, etc.).

11.1 In-process monitoring

Modern generators produce, for every cycle, a data record that typically includes:

- Peak power (W)
- Total energy (J)
- Weld time (s)
- Trigger force, weld force, hold force (N)
- Pre-trigger collapse, weld collapse, hold collapse (mm)
- Frequency excursion during the weld (Hz)
- Amplitude commanded vs. amplitude measured (μm or %)

These data are the basis of statistical process control (SPC). A part that welds with peak power 20 % above the established mean, or with a collapse 0.10 mm shorter than the mean, is statistically suspect even if it looks fine — and modern controllers allow the operator to set process limits per parameter that fault out a part automatically. Branson's Process Pulse, Dukane's iQ Explorer, Herrmann's HiQ and Telsonic's MAG controllers all include this.

11.2 Visual inspection

Visual inspection is the first line of defence. The operator and the inspector look for:

- Excess flash (over-weld) or no flash at all (under-weld).
- Discoloration, surface burning or melt streaks visible through the part wall (over-welding, over-amplitude, or cosmetic damage from the horn).
- Witness marks on the cosmetic side under the horn — "horn-mark" — which indicates excess force or over-energy.
- Mismatch / shift: the two halves are misaligned at the joint, indicating fixture or trigger problems.
- Under-collapse: the assembled height is greater than nominal — the joint did not finish.

11.3 Destructive testing

Destructive tests are run on samples (typically the first article and at established intervals during production) and on every PPAP submission:

- Tensile pull-test: the joint is loaded to failure on a universal test machine. Joint strength is recorded along with failure mode (parent material failure vs. interface failure). Interface failure indicates an under-weld; parent failure indicates a sound joint.
- Burst pressure: pressurized fluid or air is applied to the inside of a sealed part until failure. Common for medical disposables, fluid handling, and packaging.
- Cross-section: a sample weld is sectioned through the joint, polished, and examined under a microscope (10× to 100× typical). The cross-section reveals melt depth, intermixing, voids, wall flexure and other internal features that are invisible from outside. AWS G1.10M describes the inspection criteria.
- Drop test: for consumer goods, the assembled part is dropped from a specified height and inspected for joint failure.
- Vibration / fatigue: cyclic loading to a defined number of cycles (e.g., automotive: 50,000–100,000 cycles at expected service load).
- Environmental: thermal cycling (-40 °C to +85 °C, 100 cycles) followed by re-test. Reveals weak joints that pass at room temperature but fail under thermal stress.

11.4 Non-destructive testing

- Leak test (gross leak): air pressure decay or water-bath bubble check. Detects gross leaks ($>10^{-3}$ atm·cc/s).
- Leak test (fine leak): helium or a tracer gas with a mass-spectrometer; detects fine leaks down to 10^{-9} atm·cc/s. Used in medical and aerospace.
- Dye penetrant: a low-viscosity dye is applied to the joint and a developer is applied; cracks and porosity reveal as coloured spots.
- Dimensional inspection: the assembled height after weld (the most important non-destructive measurement) is checked with a height gauge or CMM. Stable height = stable weld in distance mode.
- Ultrasonic C-scan: an immersion ultrasound scan of the joint maps weld coverage. Used in aerospace composites and high-end medical.
- X-ray and CT: for high-value or safety-critical parts, micro-CT reveals internal voids and fillet shapes without sectioning.

11.5 Statistical process control plan

A typical SPC plan for a Class A ultrasonic-welded part is:

31. First-piece visual + functional check at every shift start, recipe change, or after any maintenance.
32. Run-chart of in-process data (peak power, energy, collapse) on every cycle; auto-fault on parts outside $\pm 3\sigma$.
33. Hourly leak test on a sample part for hermetic applications.

34. Daily destructive pull on 5 samples; record mean and standard deviation; plot Cpk against the spec.
35. Weekly cross-section on one part.
36. Monthly amplitude check on the horn (calibrated meter).
37. PPAP-level revalidation after any change to part design, polymer grade, mould, fixture, horn, booster, generator firmware or operator training (this list is exhaustive — any change is a change).

Capability targets

Industrial: $Cpk \geq 1.33$ on tensile strength or seal pressure. Automotive: $Cpk \geq 1.67$. Medical: $Cpk \geq 1.67$ with full traceability and ISO 13485 / 21 CFR Part 820 compliance. Aerospace and life-support medical may go to $Cpk \geq 2.0$.

12. Applicable Standards

Ultrasonic welding of plastics has its own standards lineage and inherits a number of standards from adjacent disciplines. The principal documents are summarized below.

12.1 American Welding Society (AWS) — the primary process standards

Standard	Title	Scope and use
AWS G1.1M / G1.1	Guide to Ultrasonic Assembly of Thermoplastics	The single most important reference: equipment, processes, joint design, troubleshooting, testing
AWS G1.2M / G1.2	Specification for Standardized Ultrasonic Welding Test Specimen for Thermoplastics	Test specimen geometry and tolerances for material weldability rating
AWS G1.6	Specification for the Qualification of Plastics Welding Inspectors	Inspector certification requirements (covers ultrasonic among other plastic-joining methods)
AWS G1.10M	Guide for the Evaluation of Hot Gas, Hot Gas Extrusion, and Heated Tool Butt Thermoplastic Welds	Inspection criteria including cross-section evaluation; commonly cross-referenced for ultrasonic weld evaluation
AWS B2.4	Specification for Welding Procedure and Performance Qualification for Thermoplastics	Procedure and operator qualification — the WPQR equivalent for plastic welding
AWS Handbook of Plastics Joining	(Reference handbook)	Comprehensive industry handbook covering all common plastic-joining methods

12.2 ISO and DVS — international and German standards

- ISO 472 — Plastics — Vocabulary. Defines the terminology used in plastic-joining standards.
- DVS 2216 series (German Welding Society) — Ultrasonic welding of plastics: parts cover principles, machines, joint design, testing and acoustic safety. Widely used in European OEMs and effectively the reference in German automotive.
- DVS 2202 — Imperfections in thermoplastic welds: features, descriptions, evaluations.
- ISO 13585 — Brazing — qualification test. Cross-referenced occasionally for plastic-welding terminology.

12.3 Industry-specific quality systems

- ISO 9001 — generic quality management system; the baseline.
- IATF 16949 — automotive quality management system; ultrasonic welding processes in automotive part manufacturing fall under this.
- ISO 13485 — medical device quality management; controls every aspect of medical-disposable welding from joint validation to operator certification.
- 21 CFR Part 820 — US FDA Quality System Regulation for medical devices; equivalent obligations to ISO 13485 in the United States.
- EN ISO 11553 / EN 12198 — machinery safety, applicable to the welder as a piece of machinery.
- EN ISO 13849-1 — safety-related parts of control systems; informs the safety-PLC architecture of modern welders.

12.4 Acoustic safety

- ACGIH TLV (USA) — recommends 110 dB at 20 kHz as the 8-hour TWA limit; higher for higher frequencies.
- EU Directive 2003/10/EC — workplace noise exposure limits; the principal driver for sound-enclosure design in European OEM lines.
- BS EN ISO 11688 — guidance on noise reduction at source in machinery design.

13. Top Failure Modes and How to Avoid Them

The failure modes below are listed roughly in order of decreasing frequency in production, based on the joint troubleshooting guides in AWS G1.1M Annex and the published troubleshooting tables of Branson, Dukane and Herrmann. For each failure, the symptom, root cause and corrective action are given.

#	Failure mode	Symptom	Root cause(s)	Corrective action
1	Weak weld / interface failure	Joint pulls apart at the interface, not in parent material	Insufficient energy; energy director worn or molded too small; force too low; amplitude too low; contaminated joint surface	Verify ED dimensions; raise weld limit; increase amplitude one notch; check fixture support
2	Inconsistent weld (cycle-to-cycle variation)	Strength varies across population	Worn / inconsistent ED in mold; varying material lot; loose stud joint in the stack; varying air pressure	Audit ED on CMM; lot-trace material; re-torque stud; install regulator + accumulator on air
3	Burn / discoloration / melt-through	Brown or black ring at the joint; cosmetic damage	Over-weld: too much energy or too long time; amplitude too high; trapped air igniting	Reduce weld limit; reduce amplitude; switch to distance / energy control; vent the part
4	Horn marks on cosmetic surface	Witness mark on the upper part directly under the horn	Force too high; horn face not flat or not parallel; surface texture mismatch	Reduce force; flatten / re-face the horn; add a thin protective film; use a textured horn face
5	Flash (excess melt squirting out)	Visible flash on the outside of the joint	Over-weld; ED too large; no flash trap; wall thickness insufficient at joint	Reduce weld limit; reduce ED height in next mold revision; add flash trap; thicken joint wall
6	Diaphragming / part flexure	Part visibly flexes during weld; flange cracks; no melt at the joint	Wall too thin under the horn; horn contact face not above joint; insufficient fixture support; far-field weld through compliant geometry	Move horn contact directly over joint; thicken the upper wall; add fixture support
7	No coupling / horn rattles	Loud audible noise; no melt; generator faults out	Trigger force too low; joint not parallel; horn face not in contact; cracked horn	Increase trigger force; check parallelism; inspect horn for cracks; auto-tune the stack
8	Hermetic-seal leak	Pressure decay or	ED gap (mold short-shot); polymer contamination;	Inspect ED for continuity; verify mold venting; switch to

#	Failure mode	Symptom	Root cause(s)	Corrective action
		bubble in leak test	insufficient collapse; cooling not under pressure	distance mode; extend hold time
9	Cracking near the joint	Cracks radiating from joint after weld or in service	Sharp internal corner stress concentration; resonant rib near joint; too-high amplitude on brittle polymer	Add corner radii; remove resonant features; reduce amplitude
10	Internal voids / foamed joint	Cross-section shows bubbles inside the weld; joint strength low	Hygroscopic polymer not dried (PA, PC); degraded polymer; trapped air in joint cavity	Dry the polymer; verify resin grade; vent the joint cavity
11	Horn cracking / horn failure	Visible crack at the threaded stud or at a slot; loss of amplitude	Loose stud (most common); resonance at a transverse mode; over-amplitude; abrasive material wear	Re-torque to spec; FE analysis on horn modes; replace horn; carbide-coat for abrasive resins
12	Intermittent weld faults / generator alarms	Random rejects; no clear pattern	Stack frequency drift; generator firmware issue; mains noise; air-pressure fluctuation	Stack auto-tune; firmware update; install isolation transformer; install air dryer + accumulator

The 80/20 of failures

In the field, failures #1, #2 and #3 (weak welds, inconsistency, and over-welding burn) cover roughly 80% of all production complaints. The single most useful diagnostic is to record peak power, energy and collapse on every cycle and watch for drift. Most other failure modes show up first in the trace data, before the operator notices anything visually.

14. Common Mistakes and Practical Limitations

14.1 The mistakes seen most often in plant audits

38. Designing the part first, then choosing the welding method second. Ultrasonic welding has rules — energy directors, joint geometry, wall thickness — that must be in the part from the first CAD revision. Retrofitting an energy director onto an injection-mold tool is expensive and slow.
39. Assuming "any plastic can be ultrasonically welded". Thermosets cannot be welded; semi-crystalline polymers need shear joints; flame-retarded grades may not weld at all; dissimilar polymers usually do not weld. The decision is made at material selection.
40. Specifying time-mode control for everything. Time mode does not compensate for material lot variation, mold wear or temperature drift. For new development and for any quality-critical weld, energy or distance mode is the right starting point.
41. Skimping on the fixture. A flexible or soft fixture is the most common single cause of unstable welds. The fixture must be at least as stiff as the joint it supports, and it must be supported in the joint plane.
42. Ignoring the horn's own gain when picking the booster. A horn with gain 2 plus a 1:2 booster gives 4× the converter amplitude — which on a high-amplitude generator may be far above the joint's tolerance. Always calculate total amplitude end-to-end.
43. Not drying hygroscopic polymers. Polyamide that has been on the shelf two weeks may have absorbed enough moisture to fail every weld. PC is similar. The first thing to check when a previously-stable process suddenly produces foamed joints is the moisture content of the resin.
44. Welding on cold parts. A part fresh out of injection molding (still 50–60 °C internally) welds slightly differently from a part stored overnight at 20 °C. For very stable processes either always weld within the same time window after molding, or always weld parts at room temperature.
45. Treating the 30-piece capability study as paperwork. The Cpk number is genuinely informative; if it is below 1.33 the process will produce rejects in volume production. Solving the underlying root cause is cheaper than scrapping parts in week one of production.
46. Changing one thing without re-validating. Changing the resin colourant, the regrind percentage, the operator, the fixture clamp or even the horn cleaning procedure all count as changes. Modern AIAG / IATF practice requires a documented PCN (Process Change Notice) and re-validation.
47. Cleaning the horn with abrasive pads or solvents. The horn surface texture and amplitude are part of the design. A scratched or chemically-attacked horn face changes the coupling. Clean only with a soft cloth and isopropanol; replace if texture is lost.

14.2 Practical limitations of the process

- Maximum part size: ~250 × 250 mm at 20 kHz, smaller at higher frequencies, because a single horn cannot deliver uniform amplitude over a larger area.
- Joint depth from horn (near-field vs. far-field): roughly 6 mm for amorphous, 3 mm for semi-crystalline. Beyond this, the joint becomes a far-field weld with much lower energy delivery.
- Joint thickness: very thin walls (<0.8 mm) flex during welding; very thick walls (>5 mm) absorb too much energy. The sweet spot is 1.5–3.0 mm.
- Filler content: above 30% glass or mineral, ultrasonic welding becomes marginal. Other joining methods (vibration, hot plate, laser) are usually more robust at high filler levels.
- Cycle-to-cycle thermal mass: high-volume welders running at 2–3 cycles per second develop a thermal gradient in the horn over time. Production machines often include water-cooled horns or scheduled idle periods.
- Cosmetic surfaces: ultrasonic welding always leaves a small witness mark on the upper part where the horn contacted. For ultra-cosmetic applications, the horn footprint must be invisible (hidden behind a bezel) or other welding methods must be considered.
- Part complexity: ribs, bosses, undercuts and asymmetric features near the joint disturb the energy field and cause inconsistent welds. The rule is to keep a clear region around the joint.
- Consistency over very long runs: tooling wear on the energy director is the slow-drift failure mode. Plan for it with periodic ED measurement and replaceable mold inserts.

15. Market-Leading Manufacturers and Typical Machines

The global ultrasonic welding equipment market is mature and moderately concentrated: the top five OEMs account for roughly half the market by revenue. The market is growing at a low single-digit CAGR (around 4–5%), driven primarily by automotive electrification (battery and sensor housings), medical disposables and food / pharmaceutical packaging. Below is a profile of the firms that an engineer specifying ultrasonic welding equipment in 2026 will most likely encounter.

15.1 Branson Ultrasonics (Emerson)

Branson is the historical and current market leader, and the company that effectively invented modern ultrasonic plastic welding (Robert Soloff's 1960s patents). It was acquired by Emerson Electric in 1984 and is now part of Emerson's Automation Solutions portfolio. Branson manufactures generators, stacks, presses and complete welders at facilities in the US, Mexico, Germany, the Netherlands, France, Slovakia, China, Hong Kong and Japan. The company's design guides — particularly the polymer-by-polymer joint-design tables and the Branson part-design booklet PW-3 — are the de facto industry reference. Flagship products include the GSX integrated welder series, the DCX series of digitally-controlled generators, the 2000X Actuator family, and the MicroGSX series for micro-medical assembly.

15.2 Herrmann Ultrasonics

Herrmann is a German family-owned company (founded 1961, headquartered in Karlsbad) that has grown into the second-largest ultrasonic welding OEM globally, with strong presence in European automotive, medical and battery / EV markets. Its US operation is in Bartlett, Illinois. Herrmann's HiQ DIALOG and HiQ VARIO platforms are widely used; the company is associated with the development of servo-pneumatic press technology (HMC drive system) that combines pneumatic robustness with servo-grade position control. Herrmann publishes a thorough technical library covering polymer compatibility, joint design and process tuning — its material-by-material weldability tables and the ULTRASONIC magazine archive are widely consulted.

15.3 Dukane

Dukane is a US OEM (St. Charles, Illinois) with a strong presence in plastic welding, food packaging and medical applications. Its iQ platform (iQ Servo, iQ Auto-Plus, iQ Explorer software) is a direct competitor to Branson's GSX and Herrmann's HiQ. Dukane was an early adopter of all-electric servo presses for plastics welding and its Melt-Match technology controls amplitude in real time during the cycle to compensate for changing impedance — a useful capability for tightly toleranced or hermetic welds.

15.4 Sonics & Materials

Sonics & Materials is a US company headquartered in Newtown, Connecticut. The company offers a full range of plastic and metal welders, food and laboratory ultrasonic equipment, and a large family of stacks and accessories. It is particularly well-known among medical-device contract manufacturers in North America and for the technical depth of its applications-engineering staff. The Sonics V-series press platform is a long-running, well-regarded line.

15.5 Telsonic

Telsonic is a Swiss OEM (founded 1966, headquartered in Bronschhofen) with a strong technical reputation, especially for torsional-mode ultrasonic welding (the SONIQTWIST process), which uses oscillation in the rotational rather than the longitudinal direction. Torsional welding is gentler on parts that have sensitive contents below the joint (PCBs, electronic components, optics) because there is no axial impact on the lower part — only rotational shear at the joint. Telsonic's MAG / MPX generators are widely used in European automotive and electronics.

15.6 Rinco Ultrasonics (Crest Group)

Rinco is a Swiss OEM, now part of the Crest Group (US). Strong presence in food packaging, textile / nonwoven welding (continuous ultrasonic seaming for hygiene products), and small-part medical assembly. The Rinco Standard (35 kHz) and Dynamic platforms are common in food-packaging equipment built into form-fill-seal machines.

15.7 Schunk Sonosystems

Schunk Sonosystems is a German OEM specializing more in metal ultrasonic welding (terminals, wiring harnesses, battery cell connections for EVs) than in plastics, but appears in plastic-welding market analyses because of the breadth of its presence. Worth knowing about because EV battery assemblies frequently combine metal and plastic ultrasonic welding on the same line.

15.8 Other notable suppliers

- Sonotronic Nagel (Germany) — strong in automotive and packaging.
- MS Ultraschall (Germany) — generator technology and stacks.
- Forward Technology (US, part of Crest Group) — vibration and ultrasonic welding.
- Sonobond Ultrasonics (US) — metal and plastic, with a niche in non-wovens and textile welding.
- Mecasonic (France, part of KSE Group) — European mid-market plastic welder.
- Nippon Avionics (Japan) — Japanese-market leader, strong in electronics.

- Ultrasonic Engineering (Japan), Hornwell, Maxwide, Kepu, Zhuhai Lingke (China and Korea) — large Asian suppliers serving regional volume manufacturing.

15.9 Characteristics of a typical industrial ultrasonic welder

An engineer specifying a typical 20 kHz industrial plastic welder for a small-to-medium part (e.g., a 100 mm diameter housing) in 2026 will see something close to the following specification. Numbers are representative across Branson, Herrmann, Dukane and Sonics & Materials; specific models vary in details:

Specification	Typical value
Operating frequency	20 kHz (most common); 30, 35, 40 kHz available
Generator output power	1500 – 4000 W at 20 kHz; 1000 – 2200 W at 30 / 35 kHz
Maximum horn-face amplitude	60 – 100 μ m pk-pk at 20 kHz; 25 – 45 μ m at 30 / 35 kHz
Press stroke	100 – 200 mm
Maximum down-force	1500 – 5000 N (small machine); 5000 – 12,000 N (large machine)
Trigger force range	20 – 3000 N programmable
Press drive type	Pneumatic / servo-pneumatic / all-electric servo
Position resolution (servo)	$\leq 10 \mu$ m
Force accuracy (servo)	$\pm 2\%$ of setpoint
Cycle time (typical small part)	0.3 – 1.5 s weld + 0.2 – 0.5 s hold
Control modes	Time / Energy / Distance / Peak power / Ground detect
Process data per cycle	Peak power, total energy, weld time, force profile, collapse, frequency
Connectivity	Ethernet, OPC-UA, IO-Link, MQTT for SPC and MES integration
Safety architecture	Two-hand control, light curtain or interlocked enclosure, Cat 3 / PL d safety relay
Sound enclosure	Standard for 20 kHz; optional for 30 / 35 / 40 kHz
Footprint	Approx. 0.5 – 1.0 m ² floor area for a benchtop machine; 1.5 – 3 m ² for a fully-enclosed system
Capital cost (approx.)	USD 25,000 – 45,000 entry / pneumatic; USD 60,000 – 120,000 servo with full data logging; USD 150,000+ for fully-automated cell

Selecting a welder for a new application

Pick the OEM whose installed base in your industry is large enough that local service and applications support are available within hours, not days. For volumes above a few hundred parts per shift, prefer a servo press with full data logging — the in-process data is the single best protection against quality drift, and it costs less than the warranty claims it will prevent.

16. Competing Methods and Comparative Analysis

Ultrasonic welding is one of roughly ten common techniques for joining thermoplastic parts. The right choice depends on part size, part geometry, polymer family, joint requirements, volume and budget. This section gives a brief overview of each competing method and ends with a tabular comparison.

16.1 Vibration welding (linear and orbital)

Vibration welding moves one part horizontally relative to the other at low frequency (100–250 Hz) and high amplitude (0.5–2.5 mm). The resulting frictional heat melts the joint without an energy director. Linear vibration moves in a straight line; orbital vibration moves in a small circle. Vibration welding is the workhorse for large parts (intake manifolds, instrument panels, automotive lamps) where ultrasonics cannot reach. Cycle times are typically 5–15 seconds per weld; capital cost is higher than ultrasonics.

16.2 Spin welding

Spin welding rotates one part against the other under axial force. Friction at the joint melts the polymer, the rotation is stopped at a precise angle, and the parts cool together. Spin welding is fast, strong and inexpensive — but only for parts with a circular or rotationally symmetric joint (filter housings, fluid containers, valve bodies).

16.3 Hot-plate welding

Hot-plate welding heats both joint surfaces by pressing them against a heated platen (PTFE-coated for low stick), then withdraws the platen and presses the molten faces together. Excellent for very large joints, high-strength applications and tough polymers (HDPE pipes, tanks, large automotive parts). Cycle times are 15–60 seconds because of the heat-up and changeover times. Polymer drag-out on the platen and contamination are the main process risks.

16.4 Laser welding (transmission / through-transmission infrared)

Laser welding uses a near-infrared laser (typically 808–980 nm diode or 1064 nm Nd:YAG) that passes through a transparent (laser-transmissive) upper part and is absorbed at the joint by a laser-absorbing (typically carbon-black-pigmented) lower part. The absorbed energy heats the joint, the upper part melts by conduction, and pressure from a clamping fixture closes the joint. Laser welding produces near-invisible joints, no flash, no vibration on sensitive contents, but requires careful pigment control and a clamping fixture. Used in medical (microfluidic devices), automotive electronics and high-end consumer.

16.5 Hot-gas / extrusion welding

Hot-gas welding uses a gas torch and a filler rod, like manual TIG welding for plastics. Used for fabrication and repair of large polyethylene and polypropylene tanks, ducting and pipes — not for high-volume production.

16.6 Heat staking and hot-air / cold-press

Heat staking deforms a plastic boss with a heated tool to retain a metal part (a sheet-metal cover, a PCB). It is a single-feature joining process, not a full assembly weld. The same mechanical tooling is used in cold-press / cold staking and in ultrasonic staking — the difference is the heat source.

16.7 Solvent bonding

A solvent that dissolves the polymer is applied to the joint surfaces; the parts are pressed together while the surface molecules diffuse into each other; the solvent evaporates and the joint is fused. Used almost exclusively for amorphous polymers (PMMA, PC, PS) in low-volume applications such as point-of-sale displays, signage, lab apparatus. Limited by VOC regulations in many regions.

16.8 Adhesive bonding

A liquid or paste adhesive (epoxy, acrylic, cyanoacrylate, polyurethane, silicone, hot-melt) is applied between the parts and cured. Adhesive bonding is essentially universal in materials and geometries, but introduces a third material with its own service-life, regulatory and environmental considerations. Cure times are minutes to hours; some UV-cure systems are seconds. Capital is low, per-piece cost is moderate-to-high.

16.9 Mechanical fasteners

Screws, bolts, rivets, snap-fits and threaded inserts. Mechanical fasteners are universal but they have a per-piece part cost (the fastener itself), an assembly cost (driving the fastener) and an environmental impact (steel in plastic complicates recycling). They are reversible, which is sometimes required (serviceable products). For high-volume non-serviceable assemblies, ultrasonic welding is almost always cheaper.

16.10 Comparative summary

The table on the next page summarizes the comparison across the dimensions that usually drive method selection. Ratings are qualitative (★★★★★ excellent, ★ poor) and are calibrated to a representative production application. They are not absolute and individual cases differ — a well-designed ultrasonic weld will outperform a poorly designed laser weld and vice versa.

16.11 Method comparison table

Method	Cycle time	Capital cost	Per-piece cost	Joint strength	Cosmetic / hermeticity	Max part size	Material flexibility	Best where...	Avoid where...
Ultrasonic welding	★★★★★ (0.1 – 1.5 s)	★★★★ (low–med)	★★★★★ (no consumables)	★★★★ (60–80% parent)	★★★★ (good)	★★ (≤ 250 mm)	★★★ (thermoplastics)	Small-to-medium thermoplastic parts at high volume	Large parts, thermosets, very high filler
Vibration welding (linear / orbital)	★★★ (5 – 15 s)	★★ (high)	★★★★ (no consumables)	★★★★★ (parent strength)	★★★ (some flash)	★★★★★ (large parts)	★★★★ (most thermoplastics)	Large rigid parts (intake manifolds, lamp lenses)	Very small or delicate features
Spin welding	★★★★★ (1 – 5 s)	★★★★ (low–med)	★★★★★ (no consumables)	★★★★★ (parent strength)	★★★ (visible weld line)	★★★★ (medium–large)	★★★ (most thermoplastics)	Round / circular joints (filters, fluid bottles)	Non-circular joints
Hot-plate welding	★★ (15 – 60 s)	★★★ (medium)	★★★★ (no consumables)	★★★★★ (parent strength)	★★ (heavy flash)	★★★★★ (very large)	★★★★★ (incl. tough crystalline)	Large / thick / tough crystalline (HDPE, PP tanks, pipes)	Cosmetic surfaces
Laser (TTIr) welding	★★★★★ (1 – 5 s)	★★ (high)	★★★★★ (low)	★★★★★ (very good)	★★★★★ (invisible weld)	★★★ (small–medium)	★★ (needs absorbing pigment)	Cosmetic / micro-fluidic / sensitive contents	Mass-tinted or naturally absorbing polymers
Hot-gas / extrusion welding	★ (manual, slow)	★★★★★ (very low)	★★ (filler rod)	★★★ (technique-dependent)	★ (rough)	★★★★★ (any size)	★★★ (PE, PP, PVC mostly)	Repair, fabrication, low-volume tanks and ducting	High-volume production
Solvent bonding	★★ (minutes cure)	★★★★★ (very low)	★★★★★ (low)	★★★ (moderate)	★★★★★ (clean)	★★★★★ (any)	★★ (amorphous only)	Low-volume amorphous (PMMA, PC) such as displays, signage	Semi-crystalline; regulated environments (VOC)
Adhesive bonding	★★ (cure time)	★★★★★ (very low capital)	★★ (adhesive cost)	★★★★ (good if right adhesive)	★★★★★ (clean if controlled)	★★★★★ (any)	★★★★★ (any material)	Mixed-material, complex geometries, low to medium volume	Very high volume, high temperature service
Heat staking (ultrasonic or thermal)	★★★★★ (1 – 3 s)	★★★★ (low–med)	★★★★★ (no consumables)	★★★ (sufficient for retention)	★★★ (visible boss)	★★★ (small features)	★★★★★ (most thermoplastics)	Retaining a metal or PCB component to a plastic part	Full assembly welds

Method	Cycle time	Capital cost	Per-piece cost	Joint strength	Cosmetic / hermeticity	Max part size	Material flexibility	Best where...	Avoid where...
Mechanical fasteners	★★★ (manual or automated)	★★★★★ (very low)	★ (fastener cost)	★★★★★ (very strong)	★★ (visible)	★★★★★ (any)	★★★★★ (any material)	Serviceable / repairable assemblies; high-load	High-volume disposables; recyclability

Reading the table: ultrasonic welding wins on cycle time, capital cost (compared with vibration / laser) and per-piece cost. It is competitive on joint strength and cosmetics. It loses on maximum part size (vibration and hot-plate dominate) and on cosmetic invisibility (laser wins). For mixed-material assemblies and very low volumes, adhesive bonding and mechanical fasteners remain the more flexible choice. The dominance of ultrasonic welding in high-volume thermoplastic assembly comes from being decisively better on cycle time and per-piece cost than every alternative — provided the part falls within its size and material envelope.

Decision shortcut

If your part fits in 200 × 200 mm, is made of a single thermoplastic family, has access to a horn from above, and is a high-volume product — start with ultrasonic welding. Switch to vibration welding if it is bigger, hot-plate if it is much bigger or in a tough crystalline polymer, laser if it must be cosmetically invisible, spin welding if the joint is round, and adhesive only if you cannot use any of the above.

17. References and Further Reading

17.1 Standards

- AWS G1.1M / G1.1 — Guide to Ultrasonic Assembly of Thermoplastics. American Welding Society.
- AWS G1.2M / G1.2 — Specification for Standardized Ultrasonic Welding Test Specimen for Thermoplastics.
- AWS G1.6 — Specification for the Qualification of Plastics Welding Inspectors.
- AWS G1.10M — Guide for the Evaluation of Hot Gas, Hot Gas Extrusion, and Heated Tool Butt Thermoplastic Welds.
- AWS B2.4 — Specification for Welding Procedure and Performance Qualification for Thermoplastics.
- DVS 2216 series — Ultrasonic welding of plastics (German Welding Society).
- DVS 2202 — Imperfections in thermoplastic welds.
- ISO 472 — Plastics — Vocabulary.
- ISO 9001 / IATF 16949 / ISO 13485 — Quality management systems applicable to ultrasonic welded parts in industrial, automotive and medical contexts.

17.2 Manufacturer technical literature

- Branson Ultrasonics — Plastic Joining Technical Information PW-3 (Part Design for Ultrasonic Welding); Engineered Plastics Joining brochure series; Branson Whitepapers (Process Pulse, GSX series).
- Dukane — Ultrasonic Welding Plastics: Design Guide; iQ Series technical bulletins; Melt-Match white papers.
- Herrmann Ultrasonics — Welding using ultrasonics: which plastics are suitable for welding (online technical resource); HiQ DIALOG and HiQ VARIO product literature; ULTRASONIC magazine archive.
- Sonics & Materials — Ultrasonic plastic welding application notes; V-series press technical data.
- Telsonic — SONIQTWIST torsional ultrasonic welding white papers; MAG / MPX technical data sheets.
- Rinco Ultrasonics — Standard and Dynamic platform application notes; food-packaging integration guides.

17.3 Books and handbook references

- AWS Handbook of Plastics Joining (American Welding Society).

- Grewell, D. and Benatar, A. (eds.) — *Plastics and Composites Welding Handbook*. Hanser Publishers.
- Troughton, M. J. — *Handbook of Plastics Joining: A Practical Guide*. William Andrew / Elsevier.
- Rotheiser, J. — *Joining of Plastics: Handbook for Designers and Engineers*. Hanser.

17.4 Trade media and reference articles

- Assembly Magazine — annual joining-of-plastics features; OEM round-tables on joint design.
- Plastics Technology — process and equipment reviews.
- Medical Design and Outsourcing / Medical Design Briefs — application case studies in medical plastics assembly.

17.5 Online technical resources

- Polymer compatibility databases at UL Prospector (formerly IDES) and CAMPUS — for material grade-level data.
- OEM video libraries on YouTube — particularly Branson, Herrmann, Dukane channels — for hands-on setup and troubleshooting demonstrations.
- Society of Plastics Engineers (SPE) and the SPE Joining of Plastics & Composites Division — peer-reviewed papers and conferences.

End of White Paper